

DSAA3073: Theories in Data Science

Week 9 Validation Sheet: Online Convex Optimization & Advanced Online Learning

Total Points: 130

Instructions:

- Write your answers clearly and legibly.
- Show all working for full credit.
- State any assumptions you make.
- For proofs, justify each step rigorously.
- This is a practice validation sheet. AI tools and references are allowed.
- Use this to assess your understanding before the actual test.

Part A: Multiple Choice (40 points)

Question 1. [5 marks] The Online Convex Optimization framework extends online learning to:

- a) Discrete decision spaces only
- b) Continuous convex decision spaces with convex losses
- c) Non-convex optimization problems
- d) Stochastic gradient descent

Question 2. [5 marks] In OGD, if all gradients satisfy $\|\nabla f_t(x)\| \leq G$ and the diameter of $\mathcal{K}$ is D , the optimal constant step size is:

- a) $\eta = \frac{G}{D}$
- b) $\eta = DG$
- c) $\eta = \frac{D}{G\sqrt{T}}$
- d) $\eta = \frac{\sqrt{T}}{DG}$

Question 3. [5 marks] For the simplex $\Delta_n = \{x \in \mathbb{R}^n : x_i \geq 0, \sum_i x_i = 1\}$, the Euclidean projection of $y = (2, -1, 1)$ (unnormalized):

- a) Requires solving an optimization problem
- b) Is simply $\frac{y}{\sum_i y_i}$
- c) Sets negative components to zero and renormalizes
- d) Depends on the specific loss function

Question 4. [5 marks] The telescoping argument in OGD regret analysis relies on:

- a) The potential function $\Phi_t = \sum_i w_t(i)$
- b) Tracking the squared distance $\|x_t - x^*\|^2$ which simplifies when summed
- c) Using Hoeffding's inequality
- d) The dual averaging technique

Question 5. [5 marks] FTRL (Follow-The-Regularized-Leader) with negative entropy regularizer $\psi(x) = \sum_i x_i \ln x_i$ gives:

- a) Online Gradient Descent
- b) Follow-The-Leader without regularization
- c) Exponential weights (MWU)
- d) Perceptron algorithm

Question 6. [5 marks] The regret bound for OGD has the form $\frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}$. As η increases:

- a) Both terms increase
- b) Both terms decrease
- c) First term decreases, second term increases
- d) First term increases, second term decreases

Question 7. [5 marks] Which statement about the $O(\sqrt{T})$ lower bound is TRUE?

- a) It only applies to deterministic algorithms
- b) It can be overcome with sufficient computational power
- c) It applies to all online convex optimization algorithms in adversarial settings
- d) It's specific to Euclidean geometry

Question 8. [5 marks] Projection onto the unit ball $\mathcal{K} = \{x : \|x\| \leq 1\}$ for a point y with $\|y\| > 1$:

- a) Scales y to have norm exactly 1
- b) Sets y to the origin
- c) Requires solving a quadratic program
- d) Is undefined

Part B: Mixed Format Questions (90 points)

Question 9. [15 marks] Consider OGD on $\mathcal{K} = \{x \in \mathbb{R}^2 : \|x\| \leq 1\}$ (unit ball) with $\eta = 0.2$.

Starting from $x_1 = (0, 0)$, suppose we observe gradients:

- Round 1: $\nabla f_1(x_1) = (1, 1)$
- Round 2: $\nabla f_2(x_2) = (2, -1)$

- (a) [5 marks] Compute x_2 after the first update (gradient step + projection).
- (b) [5 marks] Compute x_3 after the second update.
- (c) [5 marks] If instead $\mathcal{K} = [-1, 1]^2$ (box), recompute x_2 and x_3 using the same gradients.

Question 10. [13 marks] Prove that the Euclidean projection onto a convex set $\mathcal{K}$ is well-defined and unique.

(a) [7 marks] Show that for any $y \in \mathbb{R}^d$, the optimization problem:

$$\min_{x \in \mathcal{K}} \|x - y\|^2 \tag{11}$$

has a unique solution when $\mathcal{K}$ is closed and convex.

(b) [6 marks] Explain why uniqueness is important for OGD. What could go wrong if projection was not unique?

Question 11. [15 marks] Consider the regret bound trade-off for OGD.

- (a) [8 marks] The regret bound is $R_T(\eta) = \frac{D^2}{2\eta} + \frac{\eta G^2 T}{2}$. Plot or describe the shape of this function as η varies from 0 to infinity. Identify the optimal η by setting $\frac{dR_T}{d\eta} = 0$ and show it equals $\frac{D}{G\sqrt{T}}$.
- (b) [7 marks] Suppose $D = 1$, $G = 2$, and $T = 100$. Compute the regret at $\eta = 0.05$, $\eta = 0.1$ (optimal), and $\eta = 0.2$. What happens if we use η that's too small or too large?

Question 12. [12 marks] The connection between FTRL and MWU.

- (a) [6 marks] State the FTRL update with negative entropy regularizer $\psi(x) = \frac{1}{\eta} \sum_i x_i \ln x_i$ on the simplex Δ_n .
- (b) [6 marks] Derive the explicit form of $x_{t+1}(i)$ and show it matches the MWU exponential weights formula.

Question 13. [10 marks] (a) [5 marks] Explain why OGD uses projection while standard gradient descent does not. What role does the constraint set $\mathcal{K}$ play?

(b) [5 marks] Give an example of a constraint set where projection is computationally expensive. Why might we prefer FTRL in such cases?

Question 14. [15 marks] Synthesize the connection between online learning frameworks across Weeks 8-9.

- (a) [8 marks] Create a table or diagram showing how MWU (Week 8), OGD, and FTRL (Week 9) relate to each other. For each pair, identify if one is a special case of the other or if they share a common generalization.
- (b) [7 marks] All three achieve $O(\sqrt{T})$ regret. Explain the key mathematical property (convexity, regularization, or geometry) that enables this rate, and why it represents a fundamental limit in adversarial online learning.

